

15-Case RAG Evaluation Results

1. Purpose and scope

This document reports the evaluation of the existing RAG pipeline in `src/rag_demo.py` against 15 test questions written from the actual content of the `knowledge/` folder: 5 answerable, 5 partially answerable and 5 deliberately unanswerable. For every question it records the retrieved chunks, similarity scores, sources, final answer and whether the response was correctly grounded. It then analyses the retrieval and grounding failures found and recommends fixes. The pipeline itself was evaluated as-is and was not modified.

System under test

Corpus	14 documents in knowledge/, ingested into 163 chunks (each with a Source ID and Chunk ID)
Embeddings	sentence-transformers/all-MiniLM-L6-v2 (384 dimensions), run locally
Retrieval	Cosine similarity over all chunks; top 3 chunks passed to the model; no minimum-score threshold
Generation	Gemini (primary), with automatic fallback to Groq (openai/gpt-oss-120b) if Gemini errors
Grounding prompt	Answer only from retrieved evidence; do not invent policies, dates, fees or requirements; state when information is unavailable; do not make admission, grading or fee decisions; identify the sources used

Method and scoring

Each question was placed in `student_question` and run once with `python src/rag_demo.py` (Question 3 was run twice; see Case 3). Each response was judged on three points:

- **Retrieval:** did the top-3 chunks contain the evidence needed to answer?
- **Grounding:** is every claim in the answer supported by the retrieved text, with sources cited?
- **Result:** **Pass** = correct retrieval, grounded and appropriate answer; **Partial** = grounded but incomplete, partly inaccurate or with a small unsupported addition; **Fail** = the answer exists in the corpus but the system reported it as unavailable.

2. Results at a glance

Category	Cases	Pass	Partial	Fail	Comment
Answerable	1–5	1	1	3	Retrieval failures
Partially answerable	6–10	3	2	0	Gaps handled honestly
Unanswerable	11–15	5	0	0	All refused correctly
Total	15	9	3	3	

Key findings

- **No hallucinated university facts.** None of the 15 responses invented a policy, fee, date, name or phone number. All 5 unanswerable questions were refused, and sources were cited throughout.
- **Retrieval is the weak point, not generation.** Three of the five answerable questions failed because the chunks containing the answer never reached the model.
- **Similarity scores cannot separate answerable from unanswerable questions.** Unanswerable Case 14 scored 0.7591 at rank 1, higher than answerable Case 2 (0.6471). The pipeline relies entirely on the model to refuse.
- **Two different models produced the answers.** Gemini returned 503 “high demand” errors, so 8 of the 16 runs were answered by Groq. Grounding behaviour is therefore not compared like-for-like.

Summary of all 15 cases (top-1 chunk shown as source number-chunk number and score)

#	Category	Topic	Top-1 chunk · score	Generator	Retrieval	Grounded	Result
1	Answerable	Semester One dates	001-CHUNK-001 · 0.7603	Gemini	Correct	Yes	Partial
2	Answerable	First-year registration steps	012-CHUNK-003 · 0.6471	Gemini	Missed	Yes	Fail
3	Answerable	Pass mark and grade for 50–54%	008-CHUNK-001 · 0.7477	Groq / Gemini	Missed	Yes	Fail
4	Answerable	University Hospital services	010-CHUNK-011 · 0.7641	Gemini	Missed	Yes	Fail
5	Answerable	Graduation fees (Ugandan/East African)	009-CHUNK-010 · 0.8701	Gemini	Correct	Yes	Pass
6	Partly answerable	Registration process and deadline	012-CHUNK-033 · 0.6191	Gemini	Correct	Yes	Pass
7	Partly answerable	Timetable location and exam time	006-CHUNK-004 · 0.7014	Groq	Correct	Yes	Pass
8	Partly answerable	Accommodation cost and halls	002-CHUNK-001 · 0.7659	Gemini	Correct	Yes	Pass
9	Partly answerable	Counselling access and opening hours	005-CHUNK-002 · 0.7095	Groq	Partial	Mostly	Partial

#	Category	Topic	Top-1 chunk · score	Generator	Retrieval	Grounded	Result
10	Partly answerable	Reporting an ICT problem	013-CHUNK-004 · 0.5952	Groq	Partial	Yes	Partial
11	Unanswerable	Tuition fee, Software Engineering	009-CHUNK-009 · 0.5822	Groq	Near-miss	Yes	Pass
12	Unanswerable	Academic Registrar phone number	012-CHUNK-030 · 0.5289	Groq	Near-miss	Yes	Pass
13	Unanswerable	Dean of the School of Computing	011-CHUNK-018 · 0.3653	Groq	Irrelevant	Yes	Pass
14	Unanswerable	Personal exam timetable	006-CHUNK-004 · 0.7591	Groq	Relevant	Yes	Pass
15	Unanswerable	Personal admission status	009-CHUNK-001 · 0.6041	Gemini	Off-target	Yes	Pass

3. Test case records

Each record shows the question, the three chunks passed to the model with their similarity scores, the final answer (condensed where long), the expected answer and the assessment. Content descriptions of chunks are short summaries of the retrieved text.

i) Case 1 · Answerable · Semester One dates

Question: When does Semester One of the 2026/2027 academic year begin and end?

Generator: Gemini

Rank	Chunk ID	Score	Content
1	SRC-001-CHUNK-001	0.7603	Academic Calendar 2026/27: orientation week, semesters, exams, breaks, graduation week (one flat chunk)
2	SRC-006-CHUNK-004	0.5709	End-of-semester exam timetable is posted on college/school notice boards
3	SRC-012-CHUNK-033	0.5167	“The student registration is when” (truncated fragment)

Final answer (condensed): Semester One begins on “Saturday, August 1, 2026 (or Saturday, August 8, 2026, following Fresher’s Orientation Week)” and ends on Saturday 5 December 2026. Then states it officially runs 8 Aug to 5 Dec 2026. Source: Academic Calendar (SRC-001).

Expected: Semester One: Saturday 8 August to Saturday 5 December 2026.

Retrieval: Correct. The calendar chunk ranked first.

Grounding: Grounded and source cited.

Result: **PARTIAL** End date and final range are correct, but the answer also offers 1 August as a possible start date. That is the start of Orientation Week, not Semester One (see FL-5).

ii) Case 2 · Answerable · First-year registration steps

Question: What are the steps a first-year undergraduate student follows to complete registration?

Generator: Gemini

Rank	Chunk ID	Score	Content
1	SRC-012-CHUNK-003	0.6471	Self-registration and enrolment: students enrol each semester through the student portal
2	SRC-012-CHUNK-031	0.6150	“A student is only registered after fulfilling the following requirements:” (no items)
3	SRC-012-CHUNK-034	0.6150	Identical text to rank 2 (duplicate)

Final answer (condensed): Students must enrol each semester individually through the student portal (SRC-012). The complete, detailed steps and requirements for first-year undergraduates are unavailable in the provided evidence.

Expected: The registration requirements/steps that follow the lead-in sentence in SRC-012.

Retrieval: Missed. Two of the three slots were duplicate lead-in sentences with no list items. Ranks 4–10 were all Registration chunks (0.5192–0.6092), including CHUNK-029 (0.6092) and CHUNK-002 (0.6050), just below the top-3 cut-off.

Grounding: Grounded. The model did not invent steps.

Result: **FAIL** False “unavailable”: the answer appears to exist in the corpus but never reached the model (FL-1, FL-2).

iii) Case 3 · Answerable · Pass mark and grade for 50–54%

Question: What marks are required to obtain a Pass at Makerere University, and what grade does 50–54% receive?

Generator: Run A: Groq (Gemini returned 503). Run B: Gemini (re-run).

Run A: retrieved chunks

Ra nk	Chunk ID	Score	Content
1	SRC-008-CHUNK-001	0.747 7	“All you need to know about grading at Makerere University” (heading only, no facts)
2	SRC-009-CHUNK-001	0.658 1	Graduation Requirements introduction
3	SRC-011-CHUNK-024	0.642 5	Pass Mark and earning of credits: Pass mark is 50%

Run B: retrieved chunks

Ra nk	Chunk ID	Score	Content
1	SRC-008-CHUNK-001	0.743 8	Same grading heading chunk
2	SRC-009-CHUNK-001	0.682 9	Graduation Requirements introduction
3	SRC-011-CHUNK-001	0.624 6	Examination policy introduction (no facts)

Final answer (condensed): Run A: a credit is earned at the undergraduate Pass mark of 50% (SRC-011); the grade for 50–54% is not in the retrieved material. Run B: the sources “do not contain the specific mark ranges required to obtain a Pass or the grade assigned to marks between 50–54”.

Expected: Pass mark 50% (SRC-011-CHUNK-024). The 50–54% grade is expected in the SRC-008 grading chunks (ranked 5th and 6th; to be confirmed in knowledge/).

Retrieval: Run A got the pass mark (rank 3) but not the grade. In Run B the pass-mark chunk fell to rank 4 (0.6238) behind an intro chunk (0.6246), so neither part was retrieved.

Grounding: Grounded in both runs.

Result: FAIL The result changed between runs by a margin of 0.0008 and was never fully correct (FL-3).

iv) Case 4 · Answerable · University Hospital services

Question: What services are provided by Makerere University Hospital?

Generator: Gemini

Ra nk	Chunk ID	Score	Content
1	SRC-010-CHUNK-011	0.764 1	“University Hospital Services” (heading only)
2	SRC-010-CHUNK-012	0.687 6	“The hospital provides the following services student, staff and the general community:” (no items)
3	SRC-010-CHUNK-002	0.682 5	MakHS overview; hospital operating since 1978; training and research

Final answer (condensed): The hospital provides healthcare services to students, staff and the general community and serves as a centre for training and research. A detailed list of specific medical services is not available in the provided text (SRC-010).

Expected: The list of services that follows the lead-in sentence in SRC-010.

Retrieval: Missed. The chunk right after the lead-in, SRC-010-CHUNK-013, ranked only 10th (0.4994).

Grounding: Grounded.

Result: FAIL Same pattern as Case 2: heading and lead-in retrieved, list not (FL-1).

v) Case 5 · Answerable · Graduation fees (Ugandan/East African)

Question: How much are the graduation fees for Ugandan/East African students?

Generator: Gemini

Ra nk	Chunk ID	Score	Content
1	SRC-009-CHUNK-010	0.870 1	Fee table: convocation, graduation, certificate, transcript, gown; total UGX 200,000
2	SRC-009-CHUNK-009	0.660 4	“Graduation fees for the 76th Graduation Ceremony are as indicated below”
3	SRC-009-CHUNK-007	0.620 4	PhD graduands pay a gown fee of UGX 600,000 in addition to other fees

Final answer (condensed): A package total of UGX 200,000 with the itemised breakdown (convocation 10,000; graduation 40,000; certificate 30,000; transcript 30,000; gown 90,000) and a note on the PhD gown fee. Source: SRC-009 with URL.

Expected: UGX 200,000 package with breakdown.

Retrieval: Correct. Highest score in the whole evaluation (0.8701).

Grounding: Grounded; breakdown matches the source.

Result: **PASS** The fees are for the 76th Graduation Ceremony, while the calendar lists the 77th (January 2027). The answer does not warn that the figures may be from a previous ceremony (FL-6).

vi) Case 6 · Partially answerable · Registration process and deadline

Question: How do I register for the 2026/2027 academic year, and what is the exact registration deadline?

Generator: Gemini

Ra nk	Chunk ID	Score	Content
1	SRC-012-CHUNK-033	0.619 1	“The student registration is when” (truncated fragment)
2	SRC-012-CHUNK-003	0.559 6	Self-registration and enrolment through the student portal
3	SRC-001-CHUNK-001	0.519 2	Academic Calendar 2026/27

Final answer (condensed): Register through the student portal each semester (SRC-012). The exact deadline is not specified in the records. For reference, Semester One begins Saturday 8 August 2026 (SRC-001).

Expected: How to register (answerable); deadline (not in corpus).

Retrieval: Adequate for the question asked.

Grounding: Grounded; the deadline was correctly declined.

Result: **PASS** The detailed requirements were again not retrieved (FL-1), but the deadline part was correctly handled.

vii) Case 7 · Partially answerable · Timetable location and exam time

Question: Where can I find the university timetable, and what time is my Software Engineering exam?

Generator: Groq (Gemini returned 503)

Ra nk	Chunk ID	Score	Content
1	SRC-006-CHUNK-004	0.701 4	Exam timetable posted on college/school notice boards
2	SRC-014-CHUNK-001	0.617 7	Timetable link: timetable.mak.ac.ug
3	SRC-001-CHUNK-001	0.528 2	Academic Calendar 2026/27

Final answer (condensed): Gave the timetable portal link (SRC-014). The exam timetable is posted on college/school notice boards and students copy their personal schedule (SRC-006). No date or time for a Software Engineering exam is in the sources.

Expected: Timetable link and where exam timetables are posted; no course-specific exam time.

Retrieval: Correct.

Grounding: Grounded, sources cited.

Result: **PASS** Answerable part answered, unanswerable part declined.

viii) Case 8 · Partially answerable · Accommodation cost and halls

Question: How much does accommodation at Makerere University cost, and which halls are available for undergraduate students?

Generator: Gemini

Rank	Chunk ID	Score	Content
1	SRC-002-CHUNK-001	0.7659	Six male and three female halls of residence
2	SRC-002-CHUNK-003	0.5922	Hall attachment; nine halls accommodate 4,400 students
3	SRC-002-CHUNK-002	0.5864	Names of the nine halls

Final answer (condensed): Listed all nine halls (male: Lumumba, Livingstone, Mitchell, Nkurumah, Nsibirwa, University Hall; female: Africa, Complex, Mary Stuart). Cost information is not available in the documents (SRC-002).

Expected: Hall names (answerable); cost (not in retrieved evidence).

Retrieval: Correct for halls.

Grounding: Grounded.

Result: **PASS** To verify: SRC-002-CHUNK-009 (0.5274) and CHUNK-010 (0.5183) ranked 4th and 6th and were not read. Confirm they contain no cost information.

ix) Case 9 · Partially answerable · Counselling access and opening hours

Question: How can I get counselling at Makerere University, and what are the centre's opening hours?

Generator: Groq (Gemini returned 503)

Ra nk	Chunk ID	Score	Content
1	SRC-005- CHUNK-002	0.709 5	Purpose of the Counselling and Guidance Centre
2	SRC-005- CHUNK-006	0.673 7	“The Counselling and Guidance Centre provides several services these include:” (no items)
3	SRC-005- CHUNK-001	0.671 3	“We are here to help you address personal or emotional challenges...”

Final answer (condensed): “To obtain counselling, you can approach the centre directly...” (SRC-005). Opening hours are not in the retrieved information; suggests visiting the university website’s Counselling page or contacting the centre.

Expected: Description of the centre and its services; opening hours (not in retrieved evidence).

Retrieval: Partial. The service list (CHUNK-007, ranked 4th at 0.5644) was missed.

Grounding: Mostly grounded. “Approach the centre directly” and the website suggestion are not in the retrieved text.

Result: **PARTIAL** Combines FL-1 (list missed) and FL-6 (unsupported advice from the fallback model).

x) Case 10 · Partially answerable · Reporting an ICT problem

Question: How do I report an ICT problem, and how long will it take for the problem to be resolved?

Generator: Groq (Gemini returned 503)

Ra nk	Chunk ID	Score	Content
1	SRC-013- CHUNK-004	0.595 2	ICT knowledge base at answers.mak.ac.ug
2	SRC-012- CHUNK-012	0.304 2	Provisional registration when errors are found (irrelevant)
3	SRC-013- CHUNK-003	0.274 3	Email conflict report form

Final answer (condensed): Check the knowledge base first, and for email conflicts use the online reporting form (SRC-013). Turnaround time is not specified in the sources.

Expected: Ways to report ICT problems (knowledge base, email-conflict form, support ticket system); turnaround time (not in corpus).

Retrieval: Partial. The support-ticket chunk, SRC-013-CHUNK-001 (0.2562), ranked 5th and was missed, while an irrelevant chunk was included.

Grounding: Grounded.

Result: **PARTIAL** The answer omitted the main reporting route, the support-ticket system (FL-4).

xi) Case 11 · Unanswerable · Tuition fee, Software Engineering

Question: What is the current tuition fee for a Bachelor of Software Engineering student at Makerere University?

Generator: Groq (Gemini returned 503)

Rank	Chunk ID	Score	Content
1	SRC-009-CHUNK-009	0.5822	Graduation fees, 76th Graduation Ceremony (heading)
2	SRC-009-CHUNK-010	0.5415	Graduation fee table
3	SRC-009-CHUNK-011	0.5397	PhD cylinder fee UGX 50,000

Final answer (condensed): Declined: the retrieved information does not include tuition fees; available sources only cover graduation-related fees (SRC-009).

Expected: Refusal (tuition not in corpus).

Retrieval: Near-miss: matched on the word “fee”. “Fees and Payment” (SRC-007-CHUNK-005) ranked 8th (0.4360).

Grounding: Grounded.

Result: **PASS** Correct refusal.

xii) Case 12 · Unanswerable · Academic Registrar phone number

Question: What is the phone number of the Makerere University Academic Registrar?

Generator: Groq (Gemini returned 503)

Rank	Chunk ID	Score	Content
1	SRC-012-CHUNK-030	0.5289	All registration is handled by Registrars (process description)
2	SRC-013-CHUNK-001	0.5169	ICT support ticket system description
3	SRC-012-CHUNK-001	0.5050	Registration: college-based and self-registration

Final answer (condensed): Declined: no phone number for the Academic Registrar appears in the retrieved information; sources only describe registration and the ICT ticket system.

Expected: Refusal.

Retrieval: Near-miss (topic words matched).

Grounding: Grounded.

Result: **PASS** Correct refusal.

xiii) Case 13 · Unanswerable · Dean of the School of Computing

Question: Who will be the Dean of the School of Computing in the 2026/2027 academic year?

Generator: Groq (Gemini returned 503)

Rank	Chunk ID	Score	Content
1	SRC-011-CHUNK-018	0.3653	Deferred exam approval: “Dean or Principal” informs the Head of Department
2	SRC-010-CHUNK-008	0.3628	Biography of the hospital’s director
3	SRC-001-CHUNK-001	0.3406	Academic Calendar 2026/27

Final answer (condensed): Declined: the retrieved information does not include the name of the Dean of the School of Computing.

Expected: Refusal.

Retrieval: Irrelevant, as expected (lowest scores in the evaluation).

Grounding: Grounded.

Result: **PASS** The model did not confuse the hospital director’s biography with a dean.

xiv) Case 14 · Unanswerable · Personal exam timetable

Question: What is my personal examination timetable for Semester One 2026/2027?

Generator: Groq (Gemini returned 503)

Rank	Chunk ID	Score	Content
1	SRC-006-CHUNK-004	0.7591	Exam timetable posted on notice boards; students copy a personalised timetable
2	SRC-001-CHUNK-001	0.7087	Academic Calendar 2026/27
3	SRC-014-CHUNK-001	0.5213	Timetable link: timetable.mak.ac.ug

Final answer (condensed): Declined to provide a personal timetable. Explained that exam timetables are posted on college notice boards (SRC-006), gave the Semester One exam period (16 Nov to 5 Dec 2026, SRC-001) and the timetable portal (SRC-014).

Expected: Refusal to give personal data, with a pointer to where it can be found.

Retrieval: Relevant policy chunks retrieved.

Grounding: Grounded, sources cited.

Result: **PASS** Helpful redirection without inventing personal data. Note the top-1 score (0.7591) is higher than several answerable cases.

xv) Case 15 · Unanswerable · Personal admission status

Question: Have I personally been admitted to Makerere University for the 2026/2027 academic year?

Generator: Gemini

Rank	Chunk ID	Score	Content
1	SRC-009-CHUNK-001	0.604 1	Graduation Requirements introduction
2	SRC-012-CHUNK-001	0.492 2	Registration: college-based and self-registration
3	SRC-002-CHUNK-001	0.464 4	Accommodation: six male and three female halls

Final answer (condensed): The provided university records do not contain your personal admission status, so this information is unavailable. Sources consulted: SRC-009, SRC-012, SRC-002.

Expected: Refusal (personal record not in corpus).

Retrieval: Off-target. “Admission Lists” (SRC-003-CHUNK-001) ranked 9th (0.4216).

Grounding: Grounded.

Result: **PASS** Correct and safe, but could have pointed the student to the admission lists source.

4. Failure analysis

Six failures were identified. FL-1 to FL-4 are retrieval failures that led the model to say “unavailable” for information that appears to exist in the corpus (safe, but unhelpful). FL-5 and FL-6 are cases where the model’s answer misread or went beyond its evidence.

ID	Failure	Type	Cases	Root cause
FL-1	Lead-in sentence retrieved, its list items not	Retrieval (chunking)	2, 4, 9	Lists split from their introductory sentence; lead-in matches the query best
FL-2	Duplicate chunks fill top-k slots	Retrieval (ingestion)	2	No deduplication at ingestion
FL-3	Heading chunk outranks factual chunks; answer chunk drops out of top-3	Retrieval (ranking)	3	Short topic-dense chunks score highest; k=3 is too tight
FL-4	Relevant chunk missed, irrelevant chunk included	Retrieval (ranking)	9, 10, 15	Vocabulary gap; fixed k with no score threshold
FL-5	Conflicting Semester One start date	Grounding (generation)	1	Many dates in one flat chunk; model hedged
FL-6	Unsupported or unqualified statements	Grounding (generation)	5, 9	Fallback model added advice; dropped ceremony qualifier

FL-1: Lead-in retrieved, list items not (Cases 2, 4, 9)

What happened. For “registration steps”, “hospital services” and “counselling services”, the pipeline retrieved the sentence introducing a list but not the list itself. Case 2: SRC-012-CHUNK-031/034 “A student is only registered after fulfilling the following requirements:”. Case 4: SRC-010-CHUNK-012 “The hospital provides the following services...”, with the next chunk (CHUNK-013) ranked 10th at 0.4994. Case 9: SRC-005-CHUNK-006 “...provides several services these include:”, with the next chunk (CHUNK-007) ranked 4th at 0.5644. The model correctly said the details were unavailable.

Why. Ingestion splits text into very small chunks and separates a list from its lead-in. The lead-in repeats the question’s topic words (“services”, “requirements”), so it embeds close to the query, while bare list items do not. Only three chunks are passed on, so items ranked just below are cut off (Case 2: 0.6150 vs 0.6092).

Impact. Three cases affected, two of them full Fails: the answer exists but the student is told it does not.

Possible fix. Merge a lead-in with its list items when chunking; set a minimum chunk size; add overlap; or also retrieve the neighbouring chunk (N+1) of each top hit.

FL-2: Duplicate chunks waste top-k slots (Case 2)

What happened. SRC-012-CHUNK-031 and CHUNK-034 have identical text and identical scores (0.6150) and filled two of the three evidence slots. The truncated fragment SRC-012-CHUNK-033 (“The student registration is when”) also appeared in Cases 1, 6 and 14 as low-value evidence.

Why. The source page most likely repeats the sentence, and ingestion does not deduplicate. Truncated fragments come from the same splitting behaviour as FL-1.

Impact. Effective top-k fell from 3 to about 2 in Case 2, which made FL-1 worse.

Possible fix. Deduplicate chunks by text hash at ingestion; drop chunks below a minimum length.

FL-3: Generic heading chunk outranks factual chunks (Case 3)

What happened. For the pass-mark question the top-ranked chunk in both runs was SRC-008-CHUNK-001, “All you need to know about grading at Makerere University” (0.7477 / 0.7438), which contains no facts. Run A: the pass-mark chunk SRC-011-CHUNK-024 (0.6425) just made rank 3, so the 50% pass mark was found but not the 50–54% grade. Run B: it fell to rank 4 (0.6238) behind an intro chunk (0.6246), so neither part was answered. The grading chunks SRC-008-CHUNK-002 and CHUNK-004 ranked 5th and 6th.

Why. MiniLM rewards short, topic-dense text: section titles and intros that repeat “Makerere University”, “grading” and “requirements” score highly despite carrying no answer. Real answer chunks are longer and score lower. With k=3, a margin of 0.0008 decided whether the evidence reached the model.

Impact. The outcome changed between runs and was never fully correct for a basic grading question.

Possible fix. Attach the section title to its content chunk instead of keeping it as a separate chunk; retrieve more chunks (5–8) and rerank; add keyword (hybrid) search for literal terms such as “Pass” and “50–54%”.

FL-4: Relevant chunk missed, irrelevant chunk included (Cases 9, 10, 15)

What happened. Case 10: the support-ticket chunk SRC-013-CHUNK-001, the main route for reporting ICT problems, ranked 5th (0.2562), while top-3 included SRC-012-CHUNK-012 (0.3042), a registration-error chunk unrelated to ICT. Case 15: the admissions source SRC-003-CHUNK-001 ranked 9th (0.4216) behind graduation and registration intros, so the answer could not point the student to the admission lists. Case 9: see FL-1.

Why. The query wording (“report an ICT problem”) differs from the source wording (“support ticket”), and the embedding model handles this paraphrase gap poorly. The pipeline always passes exactly three chunks with no minimum score, so weak matches still count as evidence. In Case 10 the scores drop from 0.5952 to 0.3042, an obvious cut-off point the pipeline ignores.

Impact. Answers were correct but incomplete or less helpful; irrelevant chunks add noise to the context.

Possible fix. Relative-score cut-off (for example, drop chunks below half the top score); hybrid search; query rewriting.

FL-5: Conflicting Semester One start date (Case 1)

What happened. Retrieval was perfect (calendar chunk at rank 1, 0.7603), but the answer stated Semester One begins “Saturday, August 1, 2026 (or Saturday, August 8, 2026, following Fresher’s Orientation Week)”. 1 August is the start of Orientation Week; Semester One starts on 8 August. The correct range appeared lower in the same answer.

Why. The calendar is one flat block with eight date ranges; the first date in it is 1 August, and the model hedged instead of committing to the entry labelled “Semester One”.

Impact. A student could arrive a week early despite a correct source. This is a generation-side grounding error, not a retrieval error.

Possible fix. Split structured content (calendars, fee tables) into one entry per chunk or line; add a prompt instruction to give a single definitive answer when the evidence is unambiguous.

FL-6: Unsupported or unqualified statements (Cases 5, 9)

What happened. Case 9: the Groq answer said students can “approach the centre directly” and should visit the website’s Counselling page. Neither appears in the retrieved chunks, although the prompt forbids filling gaps with general knowledge. Case 5: the fees come from a chunk about the 76th Graduation Ceremony, while the calendar lists the 77th in January 2027; the answer presents the fees as current without saying which ceremony they apply to.

Why. For Case 9 the fallback model added natural-sounding advice. For Case 5 the model summarised the table and dropped the qualifying context.

Impact. Low to medium: unsupported advice is minor, but out-of-date fees could mislead a student.

Possible fix. Instruct the model to state the ceremony, year or effective date when quoting figures, and not to add advice beyond the evidence; re-test the fallback model against this rule.

5. Additional observations

- **Score threshold would not help.** Top-1 scores overlap across categories: unanswerable Case 14 (0.7591) and Case 15 (0.6041) versus answerable Case 2 (0.6471) and partially answerable Case 10 (0.5952).
- **Positive result.** All five unanswerable questions were refused, including Case 13, where a hospital director’s biography was retrieved for a dean question and was not misused.
- **Log label bug.** The fallback branch prints “Groq / Llama 3.3 70B”, but the model called is openai/gpt-oss-120b. This report uses the real model name.
- **Case 3 score drift.** The same question produced different scores between runs (for example SRC-011-CHUNK-024: 0.6425 vs 0.6238). With the same model and 163 chunks, scores should be almost identical, so the question wording or knowledge files may have differed between runs. Worth confirming.

6. Recommendations

These are recommendations for the next iteration; none were applied this week, since the goal was to evaluate rather than redesign the pipeline.

Priority	Recommendation	Addresses	Expected benefit
1	Fix chunking: merge headings and lead-ins with their content, set a minimum chunk size, add overlap, deduplicate by text hash	FL-1, FL-2, FL-3	Fixes the 3 outright Fails (Cases 2, 3, 4)
2	Retrieve more chunks (5–8), add a relative-score cut-off, and consider hybrid keyword + embedding search or a reranker	FL-3, FL-4	Stops answer chunks falling just outside top-3; removes irrelevant chunks
3	Split structured content (calendar, fee tables) into one entry per chunk	FL-5	Removes date ambiguity
4	Tighten the prompt: single definitive answer when unambiguous; state ceremony/year/effective date for figures; no advice beyond the evidence	FL-5, FL-6	Cleaner, safer answers
5	Fix the model label in the fallback output and log the generator used for each test	Observations	Reliable evaluation records
6	Re-run the same 15 questions after changes and compare against this baseline	All	Measurable improvement

7. Limitations

- Each question was run once; generation is not deterministic, so results could vary on re-runs.
- Only the top-3 chunk text was inspected. For chunks ranked 4–10 only the ID and score were known, so statements that a lower-ranked chunk “likely” holds the answer should be confirmed against the files in knowledge/.
- Half the runs used the fallback model, which limits comparison of grounding behaviour between cases.
- Fifteen questions give a useful diagnostic baseline, not a statistically robust measure of accuracy.